

- 27** A gene from a eukaryote is inserted into the DNA of a bacterium. The bacterium is **not** able to synthesise the polypeptide coded for by the eukaryotic gene.

Which statements explain why the bacterium is **not** able to synthesise the polypeptide?

- 1 The mRNA produced by the bacterium is different from the mRNA produced by the eukaryote.
- 2 Eukaryote genes contain non-coding sequences that are absent from bacterial genes.
- 3 There are differences in the genetic code between bacteria and eukaryotes.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 only